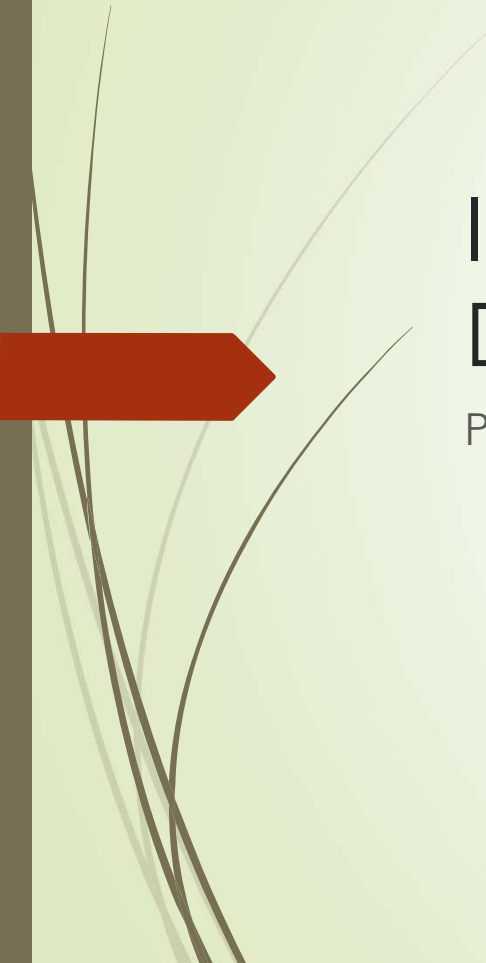




# IOT TRAINING Day-3

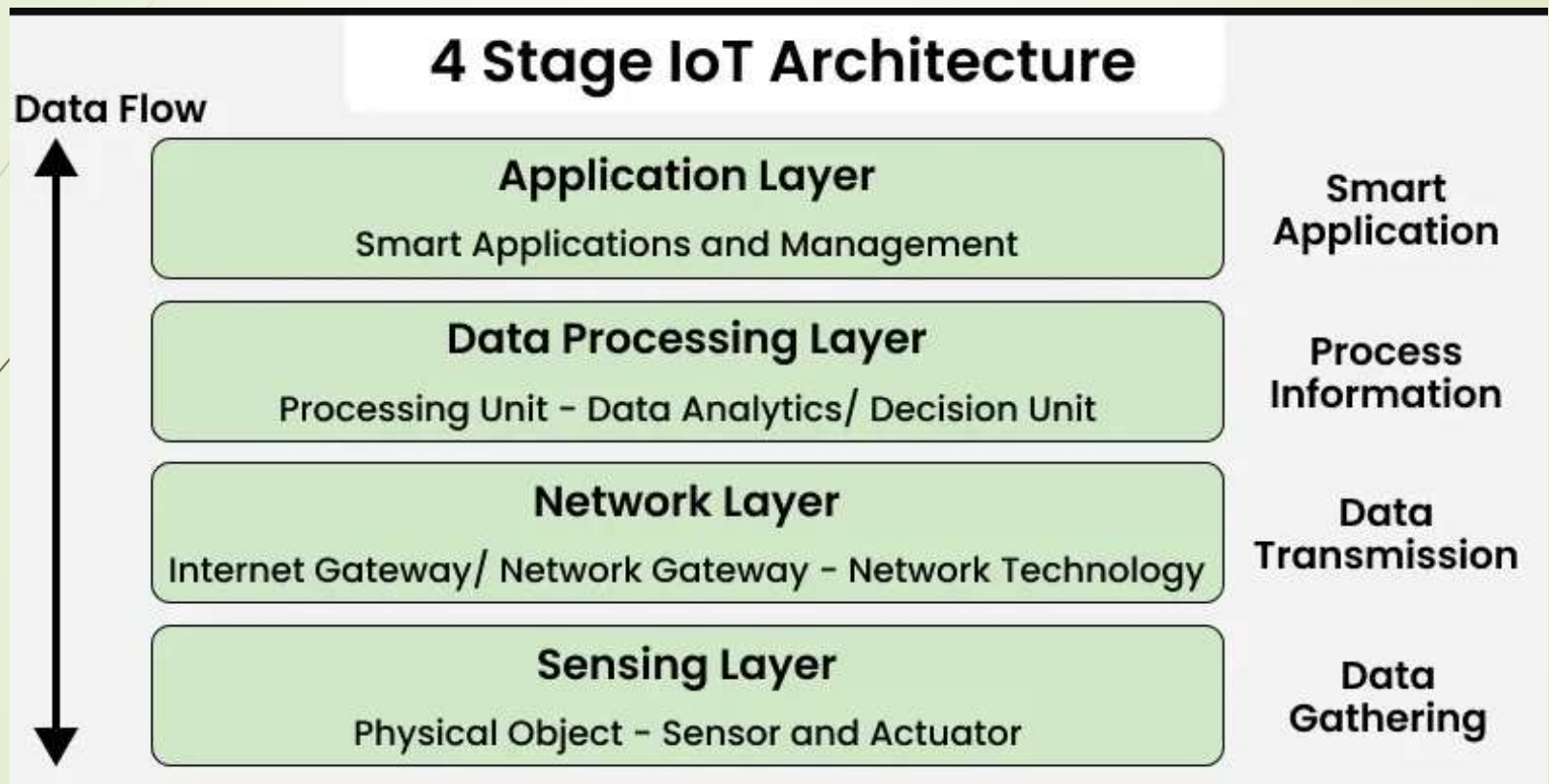
Presented by Er. Lalit Pd Joshi



## What is IOT

- The internet of things, or IoT, is a network of interrelated devices that connect and exchange data with other IoT devices and the cloud.
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# Architecture of Internet of Things





# Sensing Layer

- This is the bottom-most layer responsible for detecting physical conditions from the environment.

## Functions

- Collects raw data such as temperature, humidity, motion, sound or pressure.
- Senses changes in the surroundings through embedded components.
- Initiates actions using actuators when required.

## Components

- Sensors like humidity, gas, infrared, ultrasonic
- Actuators like motors, switches, valves
- Microcontrollers and RFID tags



# Network Layer

- This layer provides connectivity and communication between IoT devices and cloud systems.

## Functions

- Transmits collected sensor data to processing platforms securely.
- Supports device-to-device and device-to-server communication.
- Handles addressing, routing and data forwarding.

## Technologies

- Wi-Fi, Bluetooth, Zigbee, LoRaWAN
- Ethernet and satellite networks
- Supporting Devices
- Routers and switches

# Data Processing Layer

- This layer analyzes, filters and interprets data received from network devices.

## Functions

- Cleans and formats raw sensor data for meaningful insights.
- Applies analytics to detect patterns or abnormalities.
- Stores data for historical analysis or reporting.

## Components

- IoT cloud platforms
- Data lakes and warehouses
- Stream processing and machine learning engines



# Application Layer

- This top-most layer interacts directly with end users and business systems.

## Functions

- Provides interfaces to monitor and control IoT devices remotely.
- Displays visual analytics through dashboards and charts.
- Triggers automated actions based on processed insights.

## Components

- Mobile applications
- Web dashboards and portals
- Visualization and alerting tools



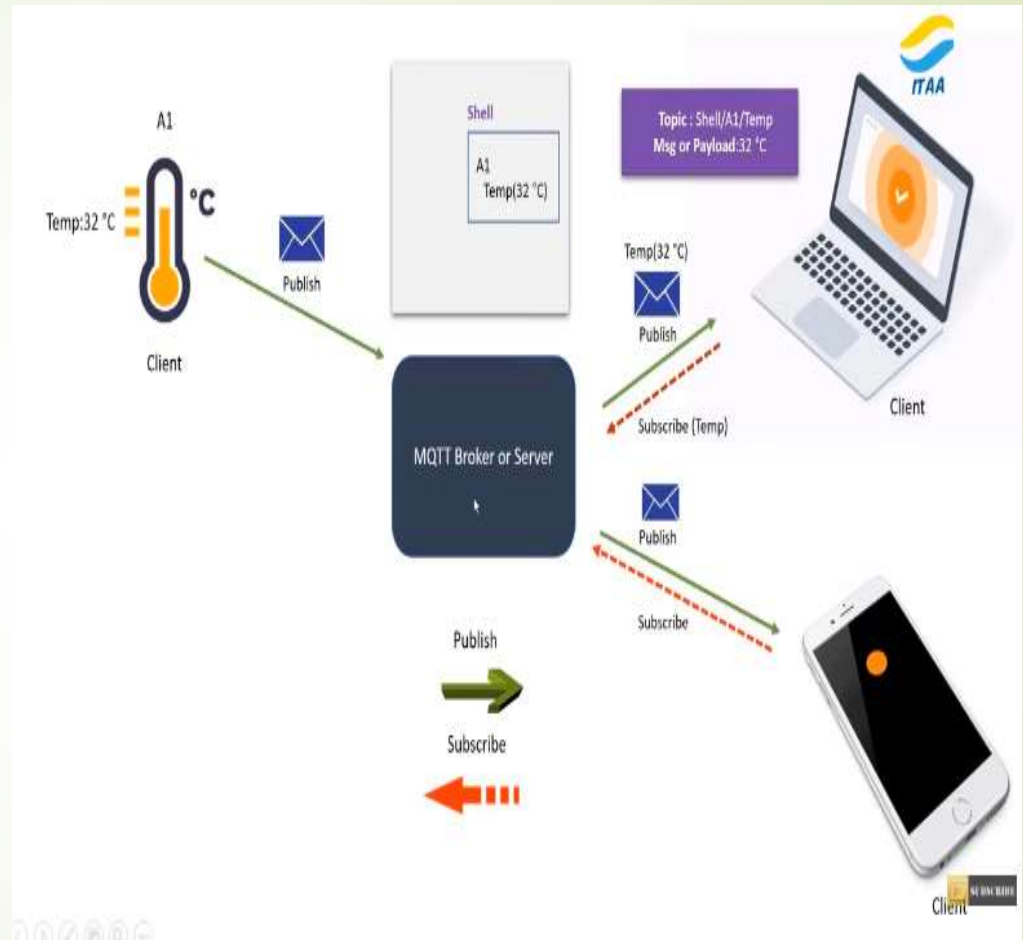
# What is MQTT

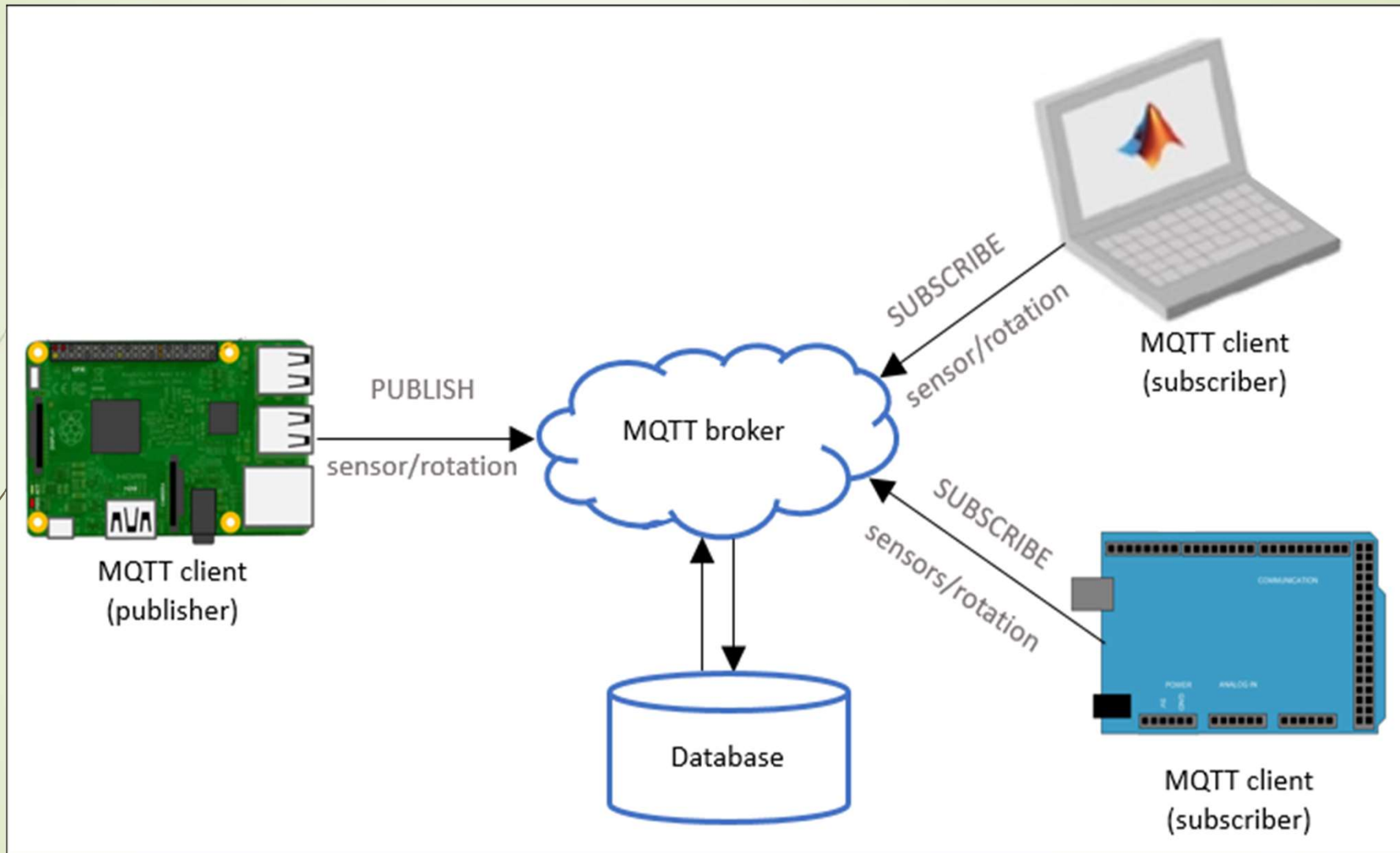
- MQTT (Message Queuing Telemetry Transport) is a lightweight, publish/subscribe messaging protocol standard for connecting IoT devices, known for being efficient on low-bandwidth, high-latency networks by using an MQTT Broker to send data from publisher to subscriber.



# Core Components

- **Client:** Any device (sensors, apps, servers) running an MQTT library, acting as a publisher (sends data) or subscriber (receives data).
- **Broker:** The central server that receives, filters, and routes messages from publishers to subscribers, managing all connections.
- **Topics:** A hierarchical string, that classifies messages, allowing clients to subscribe to specific interests.
- **Message:** The actual data payload (content) sent by a publisher, wrapped within a PUBLISH packet.







# Some modules to be installed

## **Install on Raspberry Pi**

- `sudo apt update`
- `sudo apt install mosquitto mosquitto-clients -y`

## **Enable broker at boot**

- `sudo systemctl enable mosquitto`
- `sudo systemctl start mosquito`

## **Check status**

- `systemctl status mosquitto`



## Test broker quickly

- Enter this command in one terminal
- `mosquitto_sub -h localhost -t home/test`
- Enter this command in another terminal
- `mosquitto_pub -h localhost -t home/test -m "hello"`

Notice:

You will see message coming on first terminal

## 2. MQTT Client on Raspberry Pi (Python)

- Required Python module is paho-mqtt
- `sudo apt install paho-mqtt`
- Import in Python
- `import paho.mqtt.client as mqtt`
- `sudo nano /etc/mosquitto/conf.d/listener.conf`
- Write this
  - `listener 1883 0.0.0.0`
  - `allow_anonymous true`
- `sudo systemctl restart mosquitto`



# On Esp8266

Install from **Library Manager**:

- PubSubClient(by Nick O'Leary)
- ESP8266WiFi
- #include <ESP8266WiFi.h>
- #include <PubSubClient.h>



See code on Arduino IDE



## Option A: Terminal Command (FASTEST)

- `mosquitto_pub -h localhost -t home/led -m "ON"`
- `mosquitto_pub -h localhost -t home/led -m "OFF"`